

- 4 Mitochondrial complex I is a large enzyme complex that forms part of the electron transport chain. The enzyme is composed of many different polypeptides.

The genes coding for these polypeptides are located either in mitochondrial DNA (mtDNA) or in nuclear DNA. Mutations in these genes can lead to the production of an enzyme that does not function efficiently. This results in a disease known as mitochondrial complex I deficiency. If severe, this can lead to death in early childhood.

- (a) Explain why people with mitochondrial complex I deficiency may have muscle weakness and difficulty with nervous coordination of movement.

.....

.....

.....

..... [2]

- (b) When mitochondrial complex I deficiency is caused by mutation in mtDNA:

- a cell in an ovary produces gametes with different proportions of normal mitochondria and mitochondria that contain the mtDNA mutation (mutant mitochondria)
- a person has disease symptoms when the proportion of mutant mitochondria in their cells exceeds a certain threshold
- the severity of disease symptoms, and the age at which they appear, can vary greatly in the children of one woman.

In a family with a history of mitochondrial complex I deficiency that is caused by a mutation in a **nuclear** gene, the probability of a child inheriting the mutation can be predicted.

Suggest why, in families where mitochondrial complex I deficiency is caused by mtDNA mutation, it is **not** possible to predict the probability of a child inheriting the mutation.

.....

.....

.....

.....

.....

..... [3]

- (c) Genetic screening can be carried out on people with symptoms of mitochondrial complex I deficiency.

Previously 7 mtDNA genes and 37 nuclear genes were sequenced. Some of the people tested did not have mutations in any of these genes. As a result, another gene was sequenced in these people and was found to be mutated.

This led to the suggestion that genetic screening should sequence a larger proportion of the genome for people suspected of having this disease.

Discuss the **ethical** reasons for and against sequencing a larger proportion of the genome for people suspected of having mitochondrial complex I deficiency.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (d) One mutation linked to mitochondrial complex I deficiency is a base substitution. It causes the amino acid glycine to be replaced by the amino acid valine in a region of α helix in a protein that is important for the formation of mitochondrial complex I.

Glycine is a small amino acid with an R group of one hydrogen atom whereas valine has a larger and branched R group.

Predict how the change in amino acids would affect the structure of the protein.

.....

.....

.....

..... [2]

[Total: 10]